LayerNorm Broadens Representation Geometry Without Improving Creative Surface Diversity in GPT-2

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Abstract

Creative generation often collapses toward safe and stereotyped responses, but the cause of that collapse may be architectural as well as decoding-driven. We study whether LayerNorm contributes to this effect by comparing vanilla GPT-2 against the released LN-free checkpoint LN-FREE GPT-2 under matched inference settings. Our evaluation combines 20 creative prompts from NOVELTY-BENCH and 62 validation walls from OCW, so we can measure both surface diversity and hidden-state association geometry. Removing LayerNorm increases semantic spread across repeated generations, with embedding dispersion rising from 0.803 to 0.831 ($\Delta = +0.0285$, FDR-adjusted $p = 0.024$). It also produces a much broader internal geometry on OCW: anisotropy drops from 0.992 to 0.964 ($\Delta = -0.0273$, FDR-adjusted $p = 8.5 \times 10^{-26}$) and effective rank rises from 1.35 to 2.56 ($\Delta = +1.21$, FDR-adjusted $p = 1.5 \times 10^{-11}$). However, these internal gains do not translate into reliable surface-level novelty. The LN-free model slightly reduces `distinct_1` and `distinct_2`, slightly increases lexical overlap, and does not significantly improve clue-group separability. The main conclusion is that LayerNorm compresses representation geometry in this GPT-2-family setting, but removing it alone is not enough to produce clearly more novel creative outputs.

1 Introduction

Open-ended language generation fails in two different ways. It can collapse because the decoder keeps returning high-probability continuations, and it can collapse because the model's internal geometry already favors safe associations over remote ones. The first problem is well known in text generation [Holtzman et al., 2020]. The second remains much less tested, even though recent work argues that LayerNorm changes transformer behavior at the level of residual-space projection and token selectability rather than acting only as a training stabilizer [Xiong et al., 2020, Brody et al., 2023].

Why this question matters. Creative tasks are a useful stress test because they expose the gap between the many concept combinations a model could represent and the smaller set it actually emits. If LayerNorm narrows that usable concept space, then decoding tricks alone cannot recover the lost variation. The practical implication is direct: improving creativity may require architectural interventions, not only better sampling or alignment choices.

What is missing in prior work? Existing literatures stop short of the direct test we need. LayerNorm papers explain optimization and attention geometry [Xiong et al., 2020, Brody et al., 2023, Zhu et al., 2025]. Diversity papers show that language models often underproduce varied outputs [Holtzman et al., 2020, Zhang et al., 2025]. Creativity and association benchmarks show that models can still struggle on remote association and distractor-heavy tasks [Chen and Ding, 2023,

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

Alavi Naeini et al., 2023]. However, none of these lines directly compares a normalized and LN-free language model on both creative outputs and association geometry.

Our approach. We run a matched comparison between vanilla GPT-2 and LN-FREE GPT-2, a released checkpoint with LayerNorm removed post training. We evaluate 12 generations for each of 20 creativity prompts from NOVELTYBENCH and compute lexical, semantic, and repetition-based diversity metrics. We then embed clue strings from 62 validation walls in OCW and measure association separability, anisotropy, and effective rank. This paired design lets us ask whether broader internal geometry is accompanied by broader creative behavior.

What do we find? The answer is mixed. Removing LayerNorm increases semantic response spread under fixed decoding ($0.803 \to 0.831$, FDR-adjusted $p = 0.024$), sharply reduces embedding anisotropy on OCW ($0.992 \to 0.964$, FDR-adjusted $p = 8.5 \times 10^{-26}$), and nearly doubles effective rank ($1.35 \to 2.56$, FDR-adjusted $p = 1.5 \times 10^{-11}$). But it does not deliver a reliable gain in `distinct_1`, `distinct_2`, lexical overlap, sentence adherence, or association AUC. The internal geometry broadens; the surface output does not become consistently more novel.

Our contributions. We make four concrete contributions:

- We test a direct architectural hypothesis about whether LayerNorm contributes to creative and associative collapse.
- We conduct a paired comparison that connects creative output metrics to hidden-state geometry in the same model family.
- We show that LN removal produces large, consistent geometric changes without a matching gain in lexical novelty or association separability.
- We provide a reproducible local pipeline with saved generations, wall embeddings, summary tables, and plots for follow-up analysis.

The rest of the paper reviews related work, describes the experimental protocol, presents the main results, and then discusses what the mismatch between geometry and surface diversity implies for creativity-oriented model design.

2 Related Work

LayerNorm as more than an optimization trick. Early transformer work largely treated LayerNorm as part of the recipe needed for stable training, and Xiong et al. [2020] gave one of the clearest analyses of how normalization placement affects gradients and warm-up behavior. More recent work argues that LayerNorm also changes model expressivity. Brody et al. [2023] decompose its geometric role and show that it affects which keys become selectable by attention. Zhu et al. [2025] extend the challenge further by showing that strong transformer families can work without standard normalization layers. These papers motivate our hypothesis, but they do not test whether the geometric role of LayerNorm matters for creative generation or semantic association.

Degeneration and diversity in open-ended generation. Work on neural text degeneration shows that high-probability decoding strategies can produce bland and repetitive outputs even when the underlying model is strong [Holtzman et al., 2020]. More recently, NOVELTYBENCH demonstrates that modern language models often generate far less distributional diversity than humans, even when they remain useful or coherent [Zhang et al., 2025]. This line of work frames the output-side problem clearly, but it usually studies decoding, prompting, or model scale rather than a targeted normalization ablation.

Creativity and remote association benchmarks. Creativity studies have started to measure language models with cognitive tasks rather than only subjective judgments. Chen and Ding [2023] use the Divergent Association Task to show that stochastic decoding can raise semantic-distance scores for some models, though often with a stability trade-off. Stevenson et al. [2022] evaluate GPT-3 on the Alternative Uses Test and find that humans still outperform the model on overall creative quality. Haase et al. [2025] emphasize that repeated prompting reveals large intra-model variability, which motivates our multi-sample protocol. In parallel, Alavi Naeini et al. [2023] introduce the Only Connect Wall benchmark as a distractor-rich association task that exposes brittle grouping behavior in language models. Our study builds on these benchmarks but asks a different question: whether LayerNorm changes the geometry that supports or constrains such behaviors.

Positioning our work. Unlike prior work, we do not compare unrelated model families or alternate only the decoder. We compare matched GPT-2 variants that differ in LayerNorm status, measure both external creativity and internal association geometry, and test whether the latter predicts the former. This is the missing bridge between normalization mechanics and creative-generation outcomes.

3 Methodology

Research question. Our main question is whether LayerNorm contributes to collapse toward stereotyped creative outputs by compressing the model's usable concept geometry. We evaluate three hypotheses. H1: an LN-free GPT-2 variant will generate more diverse responses under matched sampling. H2: the LN-free variant will show less collapsed hidden-state geometry, measured as lower anisotropy and higher effective rank. H3: if LayerNorm contributes to associative collapse, same-group vs. different-group clue separability should improve on OCW.

Models. We compare the Hugging Face release `gpt2` against `schaeff/gpt2-small_LNFree300`. We treat vanilla GPT-2 as the architectural baseline and LN-FREE GPT-2 as the intervention model. Both models are evaluated without additional fine-tuning or prompt engineering.

Datasets. We use two complementary benchmarks. For creative generation, we select 20 prompts from the creativity subset of NOVELTYBENCH, stored in `datasets/novelty_bench/curated.jsonl`. For association geometry, we use 62 walls from the validation split of OCW, stored in `datasets/ocw/dataset/validation.json`. The first benchmark measures output-side diversity, while the second probes representational structure on a distractor-heavy association task.

Generation protocol. For each creativity prompt and each model, we generate 12 responses with fixed sampling: temperature $= 0.9$, top-p $= 0.95$, `max_new_tokens`$= 80$, and seed $= 42$. This design follows prior diversity work in evaluating repeated generations instead of a single sample [Zhang et al., 2025, Haase et al., 2025]. For prompt-following sanity checks, we also score sentence-count adherence on the subset of prompts that explicitly requested a sentence count.

Creative-output metrics. We report `distinct_1` and `distinct_2` for lexical diversity, mean pairwise sentence-embedding cosine distance for semantic spread, mean pairwise bigram Jaccard overlap as a self-similarity proxy, and mean repetition rate as a degeneration proxy. Higher embedding dispersion indicates broader semantic exploration, while higher bigram Jaccard indicates more overlap and therefore worse diversity.

Association-geometry metrics. For each wall, we encode clue strings with the model's final hidden layer and compute four metrics. `association_auc` is the AUROC for classifying same-group vs. different-group clue pairs from cosine similarity. `similarity_gap` is mean within-group minus between-group similarity. `anisotropy` is mean pairwise cosine among clue embeddings. `effective_rank` is a participation-ratio estimate of covariance dimensionality. Under our hypothesis, lower anisotropy and higher effective rank indicate less representational collapse.

Statistical testing. The unit of analysis is the prompt for generation metrics and the wall for geometry metrics. For each paired comparison, we first run Shapiro–Wilk on the paired differences. If normality is not rejected, we use a paired t-test; otherwise we use the Wilcoxon signed-rank test. All tests are two-sided. We then apply Benjamini–Hochberg false discovery rate correction within each metric family. We report effect sizes from the saved analysis pipeline and include 95% bootstrap confidence intervals for model means.

Implementation details. The study runs with Python 3.12.8, `torch 2.11.0+cu130`, `transformers 5.7.0`, `sentence-transformers 5.4.1`, `scikit-learn 1.8.0`, `scipy 1.17.1`, `pandas 3.0.2`, `matplotlib 3.10.9`, and `seaborn 0.13.2`. The workspace exposed four NVIDIA RTX A6000 GPUs with 49 GB memory each, but CUDA was unavailable because the installed Torch build expected a newer driver than the container provided. As a result, all experiments ran on CPU and completed in 262.65 seconds. The main script is `src/run_layernorm_study.py`, with raw generations and metadata saved under `results/layernorm_study/`.

Metric	GPT-2	LN-FREE GPT-2	Delta (LN-free − GPT-2)	FDR-adjusted p
distinct_1	**0.463** [0.449, 0.476]	0.451 [0.439, 0.464]	-0.0117	0.258
distinct_2	**0.899** [0.888, 0.909]	0.884 [0.871, 0.896]	-0.0151	0.115
embedding_dispersion	0.803 [0.772, 0.833]	**0.831** [0.804, 0.857]	+0.0285	**0.024**
pairwise_bigram_jaccard↓	**0.0063** [0.0053, 0.0074]	0.0081 [0.0067, 0.0096]	+0.0018	0.105
mean_repetition_rate↓	0.0277 [0.0170, 0.0404]	**0.0256** [0.0168, 0.0365]	-0.0021	0.766
sentence_adherence	**0.222** [0.167, 0.250]	0.167 [0.083, 0.250]	-0.0556	0.600

Table 1: Creative-generation results on 20 NOVELTYBENCH prompts. Entries report mean with 95% bootstrap confidence interval in brackets. For pairwise_bigram_jaccard and mean_repetition_rate, lower is better; higher is better for the other metrics. Best values are in **bold**.

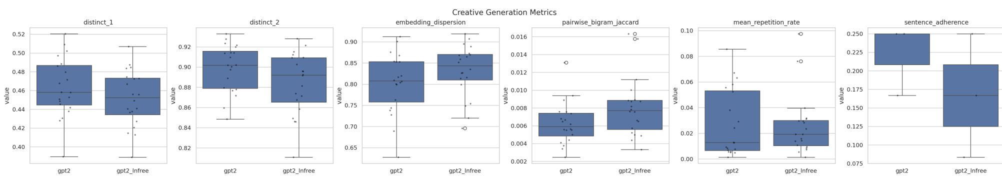

Figure 1: Creative-generation metrics across 20 NOVELTYBENCH prompts. The LN-free model increases semantic dispersion, but the lexical metrics do not improve in parallel. This separation between semantic spread and surface distinctness is the central result on the generation side.

4 Results

Creative generation. Table 1 shows the full prompt-level comparison on NOVELTYBENCH. The only result that survives FDR correction is semantic spread: embedding dispersion rises from 0.803 to 0.831, with a moderate paired effect size ($d = 0.73$). In contrast, the LN-free model slightly lowers distinct_1 and distinct_2, slightly raises bigram overlap, and leaves repetition and sentence adherence effectively unchanged. The directional checks match this mixed picture. LN-FREE GPT-2 shows higher semantic dispersion on 15 of 20 prompts, but it beats GPT-2 on only 8 of 20 prompts for distinct_1 and 6 of 20 prompts for distinct_2. Figure 1 visualizes the same pattern: broader semantic spread without cleaner lexical novelty.

Association geometry. Table 2 shows a much stronger effect on the OCW benchmark. Removing LayerNorm does not significantly improve association AUC or similarity gap, but it changes embedding geometry substantially. Mean anisotropy falls from 0.992 to 0.964, and effective rank rises from 1.35 to 2.56. Both survive FDR correction by a large margin. The directional checks are even stronger: LN-FREE GPT-2 has lower anisotropy on all 62 walls and higher effective rank on all 62 walls. However, AUC improves on only 33 of 62 walls, which means the broader geometry does not translate into a reliable gain in clue-group separability.

Statistical interpretation. Taken together, the results support H2 strongly, support H1 only through semantic dispersion, and do not support H3 strongly enough to claim better association structure at the task level. The key empirical fact is therefore not a broad creativity gain, but a mismatch: LayerNorm removal expands representational geometry much more than it improves surface behavior.

5 Discussion

What do the results mean? The cleanest conclusion is that LayerNorm acts as a geometry-shaping component in GPT-2-class models. On OCW, removing it produces less anisotropic and higher-rank clue embeddings for every wall in the evaluation set. This is exactly the direction expected if LayerNorm helps compress the residual stream into a more homogeneous geometry. The generation results partially agree with that story: semantic spread increases under fixed decoding, which suggests that the model explores a broader region of concept space when LayerNorm is absent.

Why does broader geometry not become broader lexical novelty? Our results suggest that representation breadth is necessary but not sufficient. The LN-free model may access more varied hidden states while still decoding into similar lexical forms because the pretrained distribution, model capacity, and weak prompt following of GPT-2 remain the dominant bottlenecks. In other words, a

Metric	GPT-2	LN-FREE GPT-2	Delta (LN-free − GPT-2)	FDR-adjusted p
`association_auc`	0.527 [0.515, 0.540]	**0.529** [0.519, 0.541]	+0.0023	0.706
`similarity_gap`	0.00058 [0.00031, 0.00089]	**0.00105** [0.00024, 0.00186]	+0.00047	0.299
`anisotropy`$\downarrow$	0.992 [0.991, 0.992]	**0.964** [0.961, 0.967]	-0.0273	8.5×10^{-26}
`effective_rank`	1.35 [1.27, 1.45]	**2.56** [2.35, 2.79]	+1.21	1.5×10^{-11}

Table 2: Association-geometry results on 62 OCW validation walls. Entries report mean with 95% bootstrap confidence interval in brackets. Lower anisotropy is better; higher is better for the remaining metrics. Best values are in **bold**.

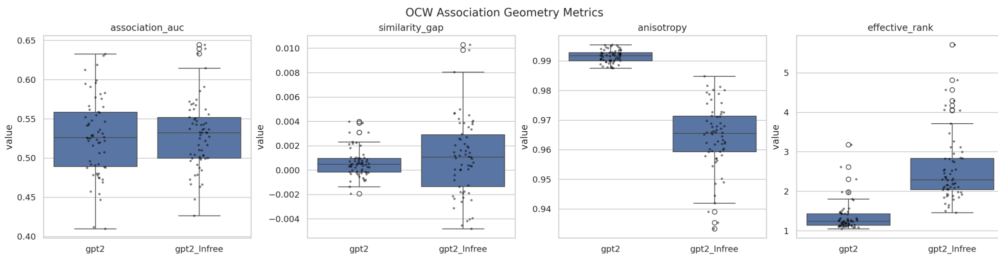

Figure 2: Association-geometry metrics across 62 OCW validation walls. LN removal consistently lowers anisotropy and raises effective rank, but the separability metrics remain close to baseline. The figure highlights a strong internal change without a matching task-level gain.

broader internal manifold does not automatically become better controlled creativity at the output surface.

Alternative explanations. One possibility is that LN-FREE GPT-2 is simply noisier after post-training LayerNorm removal. That could increase embedding dispersion without improving useful novelty. A second possibility is benchmark mismatch: NOVELTYBENCH was designed for stronger instruction-following models, so GPT-2 may be too weak for lexical-diversity gains to emerge cleanly. A third possibility is measurement mismatch. Our OCW probe uses direct clue embeddings from a causal language model rather than a task-trained grouping head, so it may reveal geometry more readily than task competence.

Limitations. This study uses a single model family, a small released LN-free derivative, and only one normalization ablation. The sentence-adherence metric is based on just three prompts and is therefore too sparse to carry much weight. The geometry benchmark is indirect, because we evaluate clue embeddings rather than a supervised association model. Finally, the experiments ran on CPU after a CUDA driver mismatch, which did not affect correctness but limited the scope of possible follow-up sweeps.

Broader implications. These findings narrow the claim we can make. They do not show that LayerNorm is the main cause of creative collapse in modern aligned language models. They do show that normalization changes internal geometry in a direction consistent with representational compression, and that removing it alone does not solve the harder problem of turning latent variation into controlled, useful novelty. For creativity-oriented model design, that points toward combined interventions: architecture, decoder, and evaluation must likely move together.

6 Conclusion

We examined whether LayerNorm contributes to creative and associative collapse by comparing GPT-2 with a matched LN-free checkpoint across creative generation and association geometry. The answer is partial support for the hypothesis. Removing LayerNorm strongly broadens hidden-state geometry, as shown by large reductions in anisotropy and large increases in effective rank, and it also raises semantic spread across repeated generations. However, it does not yield a reliable gain in lexical distinctness, prompt adherence, or clue-group separability.

The main takeaway is simple: in this GPT-2-family setting, LayerNorm looks like a geometry-shaping factor rather than a single direct cause of creative-output collapse. Future work should test stronger model families with matched normalized and de-normalized variants, add human or strong-

LLM quality judgments to separate novelty from incoherence, and study architecture-by-decoder interactions directly under greedy, temperature, and nucleus sampling.

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